



Project Title : Movie and TV series info system

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Abstract

This project aims to develop a comprehensive database system for storing and managing information about movies and TV series. The database captures details such as titles, genres, casts, release dates, runtime, ratings, and reviews. It provides a structured way to store data efficiently, supports queries for retrieving information, and allows for future enhancements such as recommendation systems or integration with streaming platforms.

Introduction

The entertainment industry generates vast amounts of data daily, including movies, TV series, cast members, and user reviews. Managing this data manually is inefficient and error-prone. This project focuses on creating a relational database system to store and manage such information efficiently. The database supports queries for retrieving media details, cast information, genre classifications, and ratings.

Objectives

- To store detailed information about movies and TV series.
- To maintain relationships between media, genres, and cast members.
- To support data retrieval through structured queries.
- To enable future integration with applications such as streaming platforms or recommendation engines.

Requirement Analysis

Functional Requirements

- Store information about media (movies/TV series).
- Store information about genres.
- Maintain cast and crew information.
- Allow querying media by release date, genre, or cast.
- Store ratings and reviews.

Non-Functional Requirements

- Data integrity through constraints and relationships.
- Efficient query execution.
- Scalability for large datasets.

Database Design

The database uses a relational model with the following primary entities:

1. Media: Stores general information about movies and TV series.
2. Genre: Stores genre names and descriptions.
3. Media_Genre: Many-to-many relationship between media and genres.

4. Cast: Stores actors, directors, and other contributors.
5. Media_Cast: Many-to-many relationship between media and cast members.

Schema Description

Media Table

Column	Type	Constraints
media_id	NUMBER	PRIMARY KEY
title	VARCHAR2(255)	NOT NULL
overview	CLOB	
release_date	DATE	
runtime	NUMBER	
media_type	VARCHAR2(10)	NOT NULL

Genre Table

Column	Type	Constraints
genre_id	NUMBER	PRIMARY KEY
name	VARCHAR2(50)	NOT NULL, UNIQUE

Media_Genre Table

Column	Type	Constraints
media_id	NUMBER	FOREIGN KEY REFERENCES Media(media_id)
genre_id	NUMBER	FOREIGN KEY REFERENCES Genre(genre_id)

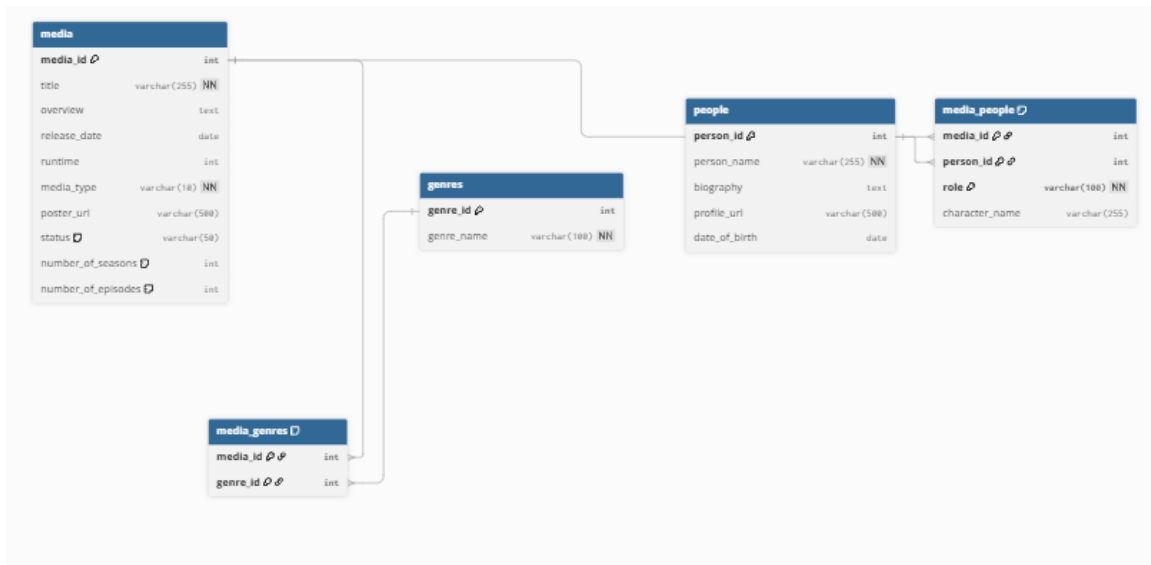
Cast Table

Column	Type	Constraints
cast_id	NUMBER	PRIMARY KEY
name	VARCHAR2(255)	NOT NULL
role	VARCHAR2(100)	NOT NULL

Media_Cast Table

Column	Type	Constraints
media_id	NUMBER	FOREIGN KEY REFERENCES Media(media_id)
cast_id	NUMBER	FOREIGN KEY REFERENCES Cast(cast_id)

Schema Diagram



ER Diagram

